Keith J. Polson
Site Vice President

DTE Energy Company
6400 N. Dixie Highway, Newport, MI 48166
Tel: 734.586.4849 Fax: 734.586.4172
Email: keith.polson@dteenergy.com

10 CFR 50.73

March 6, 2017
NRC-17-0018

U. S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555-0009

Reference: Fermi 2
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2017-001

Pursuant to 10 CFR 50.73(a)(2)(v)(A) and (D), DTE Electric Company (DTE) is submitting LER No. 2017-001, Loss of Reactor Protection System Scram Function During the Main Steam Isolation Valve and Turbine Stop Valve Channel Functional Tests due to Use of a Test Box.

No new commitments are being made in this LER.

Should you have any questions or require additional information, please contact Mr. Scott A. Maglio, Manager – Nuclear Licensing, at (734) 586-5076.

Sincerely,

Keith J. Polson
Site Vice President

Enclosure: Licensee Event Report No. 2017-001

cc: NRC Project Manager
NRC Resident Office
Reactor Projects Chief, Branch 5, Region III
Regional Administrator, Region III
Michigan Public Service Commission
Regulated Energy Division (kindschl@michigan.gov)

Enclosure to
NRC-17-0018

Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43

Licensee Event Report (LER) No. 2017-001

NRC FORM 366 (06-2016)

U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 10/31/2018

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

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4. TITLE
Loss of Reactor Protection System Scram Function During Main Steam Isolation Valve and Turbine Stop Valve Channel Functional Tests due to Use of a Test Box

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
01	06	2017	2017	- 001 -	00	03	06	2017	N/A	05000
									N/A	05000

9. OPERATING MODE: 1

10. POWER LEVEL: 100

11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☑ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☑ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER (Include Area Code)
Fermi 2 / Scott A. Maglio – Manager, Nuclear Licensing	(734) 586-5076

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☑ NO				

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On January 6, 2017 an Operations Shift Engineer determined that use of the Reactor Protection System (RPS) test box described in station procedures would result in the loss of two RPS reactor scram functions. Technical Specification (TS) 3.3.1.1 requires that RPS instrumentation for Table 3.3.1.1-1 Function 5 for Main Steam Isolation Valves (MSIVs) and Table 3.3.1.1-1 Function 9 for Turbine Stop Valves (TSV) remain OPERABLE. Operations procedures were revised to incorporate the use of the test box in August of 2016. Between September 22 and 23, 2016 the MSIV and TSV procedures were each performed one time using the test box. The failure to recognize the impact of the procedure revisions is considered a human performance error by engineering and operations personnel.

The procedures were corrected in January 2017 to remove the use of the RPS test box. Subsequently, on January 7 and 9, 2017, respectively, the procedures for the TSVs and the MSIVs were performed successfully.

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NARRATIVE

INITIAL PLANT CONDITIONS

Mode – 1
Reactor Power – 100 percent

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

On January 6, 2017 an Operations Shift Engineer (i.e. a Licensed Operator) determined that use of the Reactor Protection System (RPS) [JC] test box [IIS] in procedures 24.137.0.1 "Main Steam Line Isolation Channel Functional Test" and 24.110.05, "Turbine Control and Stop Valve Functional Test" would result in the loss of RPS reactor scram functions. Technical Specification (TS) 3.3.1.1 requires that RPS instrumentation for Table 3.3.1.1-1 Function 5 "Main Steam Isolation Valve – Closure" for Main Steam Isolation Valves (MSIVs) [ISV] and Table 3.3.1.1-1 Function 9 "Turbine Stop Valve – Closure" for Turbine Stop Valves (TSV) [ISV] remain OPERABLE.

Both Operations procedures were revised to incorporate the use of a test box in August of 2016. Implementation of the test box was intended to reduce unnecessary RPS actuations by eliminating the half scram created during previous test procedure performances. Between September 22 and 23, 2016, the MSIV and TSV procedures were each performed one time using the test box. The performance of the MSIV and TSV procedures using the test box caused the loss of the RPS trip functions by bypassing more than the TS minimum allowed inputs per trip channel to maintain trip function operability.

The unintended loss of RPS trip functions during these tests resulted in a NRC reportable condition under 10CFR50.73(a)(2)(v) as "any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to: 10CFR50.73(a)(2)(v)(A) shutdown the reactor and maintain it in a safe shutdown condition," and 10CFR50.73(a)(2)(v)(D) "Mitigate the consequences of an accident."

The error in the procedures was self-identified by the Operations Shift Engineer during a procedure review on January 6, 2017. The procedures were corrected reinstating the previous method of testing such that the test box was no longer used. Subsequently, on January 7 and 9, 2017, respectively, the procedures for the TSVs and the MSIVs were performed successfully.

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

The test box was designed to reduce unnecessary RPS actuations. The test box consists of a 3-ohm resistor in parallel with a 5 VAC lamp terminated with banana jacks. The RPS test box is a low resistance path in parallel with the trip logic relay contacts.

The RPS initiates a reactor scram when at least one MSIV in 3 of 4 Main Steam Lines (MSLs) close (Function 5) or when 3 of 4 TSVs close (Function 9). The automatic MSIV and TSV closure reactor scrams preserve the integrity of the fuel cladding and the Reactor Coolant System (RCS) in anticipation of the transients caused by closure of these valves.

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The RPS has two independent trip systems (A and B) with two logic channels in each trip system: channels Al and A2, B1 and B2. The use of the RPS test box, as implemented in the procedures at the time of their use in September 2016, would bypass valve position inputs for one trip logic channel preventing the trip logic channel from being in a tripped condition (half-scram). For Function 5, the logic used ensures a full reactor scram occurs for the condition where at least one MSIV in three or more MSLs are less than 90% open. This occurs as the logic uses valves A and B for the A1 trip logic, C and D for the A2 trip logic, A and C for the B1 trip logic, and B and D for the B2 trip logic. For Function 9, the logic used ensures a full reactor scram occurs for the condition where three or more TSVs are less than 90% open. The use of the RPS test box, as implemented in September 2016, would still result in a half-scram for the trip system under test because each trip logic channel (A1, A2, B1, B2) individually produces a half-scram. Thus, the entire logic would have remained capable of initiating a full reactor scram. However, the TS 3.3.1.1 Bases requirement of 3 valve position signals per trip system was temporarily (for the duration of test) not met.

Technical Specification 3.3.1.1 Condition C requires restoring RPS trip capability. For Function 5, this would require both trip systems to have each channel associated with the MSIVs in three main steam lines (not necessarily the same main steam lines for both trip systems) OPERABLE or in trip (or the associated trip system in trip). For Function 9, this would require both trip systems to have three channels representing three TSVs, each OPERABLE or in trip (or the associated trip system in trip).

The required action and completion time for TS 3.3.1.1 CONDITION C:

C. REQUIRED ACTION – Restore RPS trip capability, COMPLETION TIME – 1 hour

Condition C was applicable to both the MSIV and TSV RPS logic functional testing.

The longest time the test box was installed during the performance surveillances was 55 minutes for the MSIV surveillance and 57 minutes for the TSV surveillance. TS 3.3.1.1 Required Action C was not intentionally entered during the performance of the surveillance but was met each time.

The surveillance procedural error was identified by the Operations Shift Engineer in January 2017 during the work control review for the next scheduled surveillance. Procedures 24.137.01 and 24.110.05 were subsequently revised to not use the test box for future surveillance tests.

Each surveillance procedure was performed once in September 2016 before the error was recognized in January 2017. The procedures were revised and subsequently performed correctly during the next surveillance test in January 2017.

CAUSE OF THE EVENT

Procedures 24.137.01 and 24.110.05 were revised in August of 2016, to include use of the test box. Neither the procedure Technical Review nor the 10 CFR 50.59 evaluation recognized that use of the RPS test box in procedures 24.137.01 (MSIV) and 24.110.05 (TSV), resulted in bypassing the parallel contacts of multiple trip relays associated with the MSIVs or TSVs closure. In the test condition, the number of operable channels in both MSIV and TSV RPS trip systems was reduced such that the RPS trip capability was not maintained as described by the TS 3.3.1.1 Bases.

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NARRATIVE

While the requirement of having 3 valve position signals per trip system to maintain the RPS trip capability was recognized during preparation of the 10 CFR 50.59 evaluation, the understanding of how this requirement translates to relay contact logic development and the complexities introduced by the parallel logic strings was not identified. The failure to recognize the unique attributes of this logic and the impact of the procedure revisions are considered a human performance error by the engineering and operations staff.

CORRECTIVE ACTIONS

The procedures 24.137.01 and 24.110.05 were revised to remove the use of the test box during future surveillance tests. Subsequently, on January 7 and 9, 2017, respectively, the procedures for the TSVs and the MSIVs were performed successfully without the use of the test box.

A site wide Human Performance Event reset was conducted to communicate lessons learned from this event.

Additional coaching will be performed for Engineering and Operations personnel on lessons learned when conducting procedure changes such as: (1) supporting applicable Licensing Basis documents need to be reviewed, (2) supporting evaluations need to be adequately intrusive and to the required depth, (3) risk needs to be articulated, and (4) cross organizational expertise needs to be proactively sought before proceeding with the procedure change.

In addition, project plans will be modified to communicate and implement lessons learned for a planned future application of a MSIV isolation logic test box.

PREVIOUS OCCURRENCES

None.